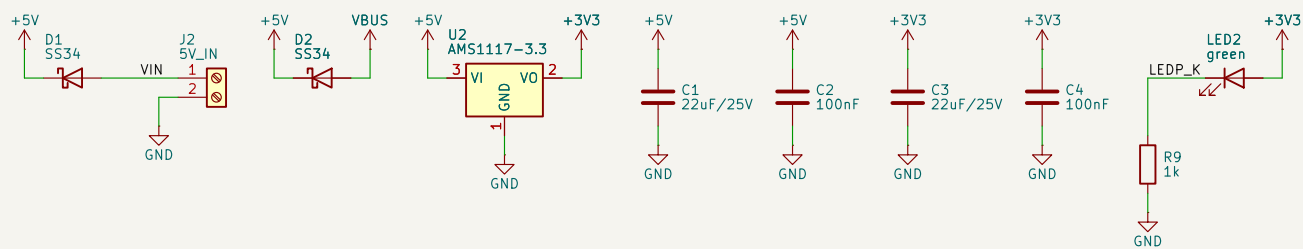
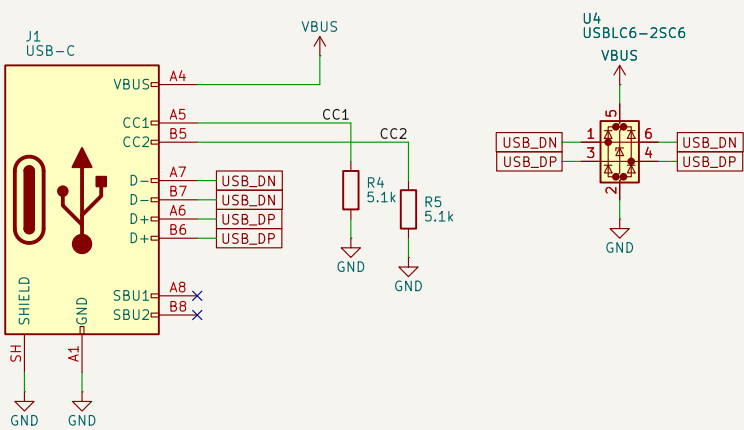


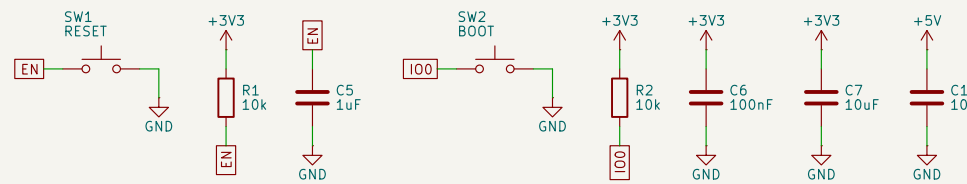
POWER IN  
5V DC terminal or USB, ORed Schottky diodes



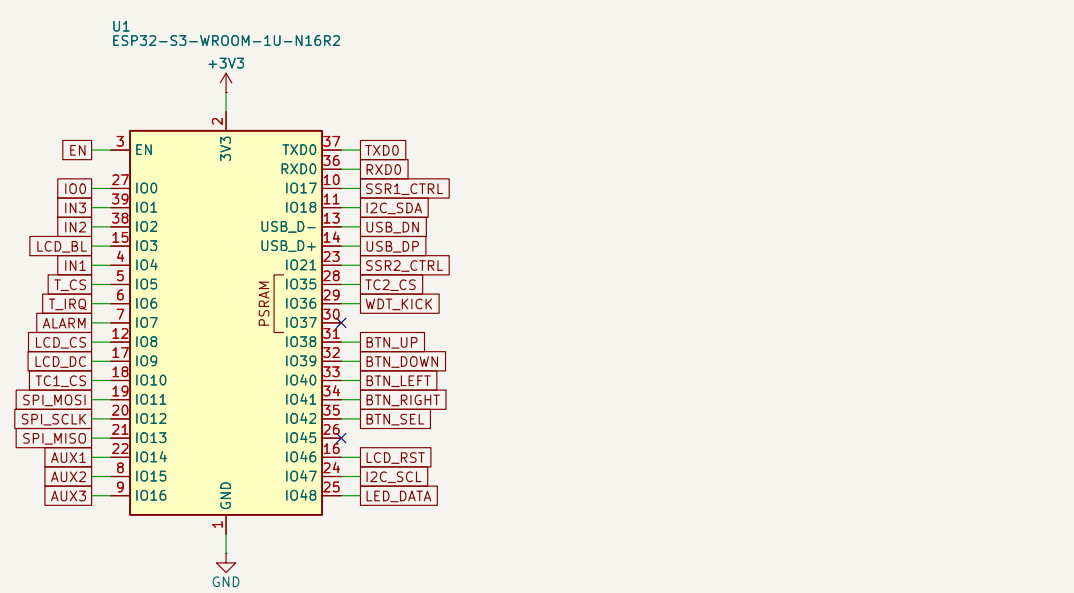
USB-C  
native USB flashing + ESD



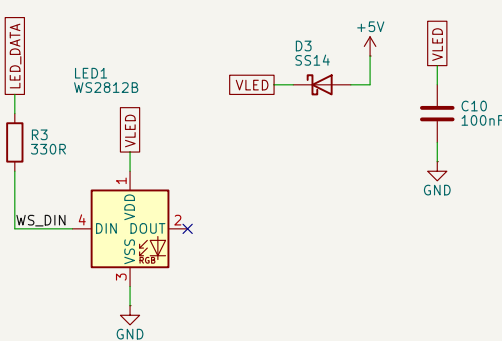
RESET / BOOT / DECOUPLING



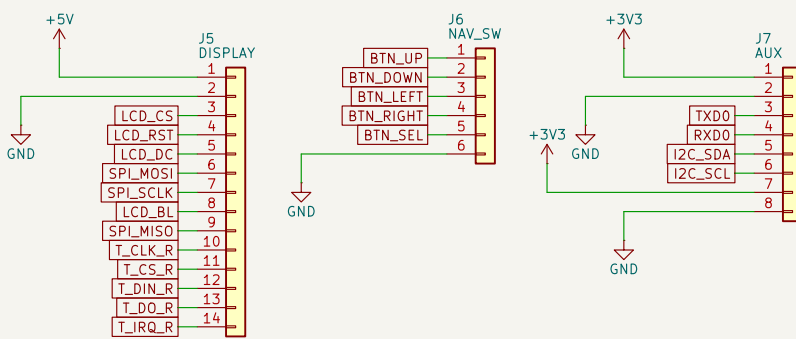
ESP32-S3-WROOM-1U-N16R2  
GPIOs = firmware Kconfig defaults



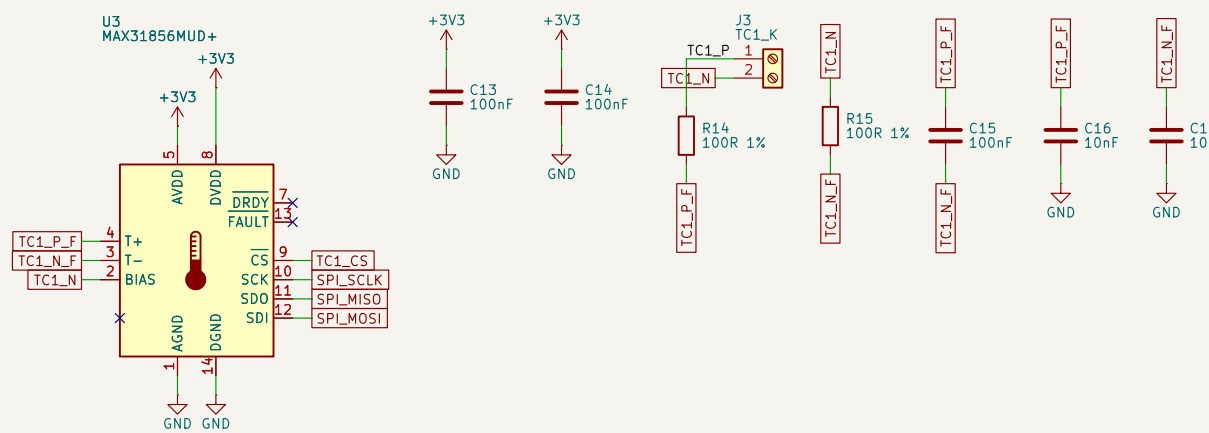
WS2812B STATUS LED  
VDD dropped -4.6V for 3.3V data margin



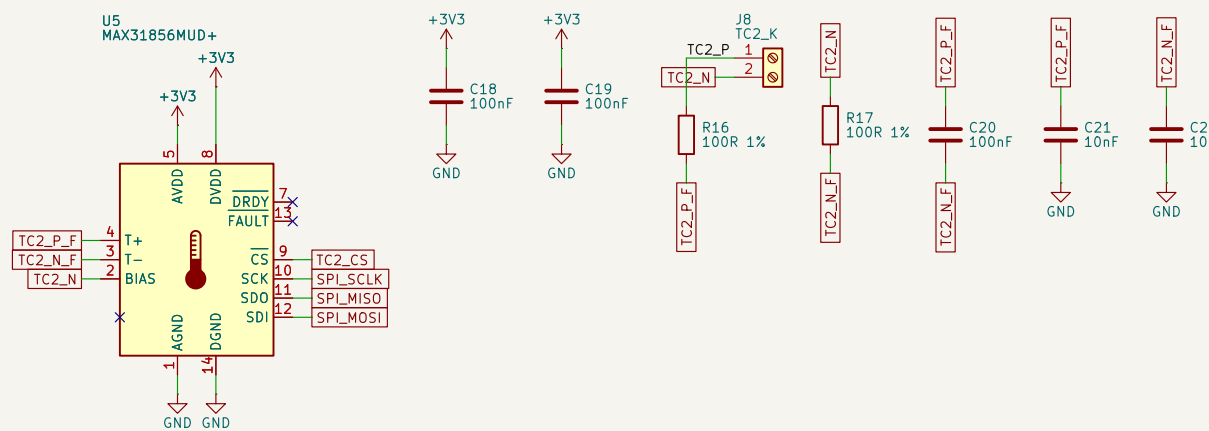
HEADERS  
J5 display+touch (14-pin), J6 nav, J7 aux+I2C



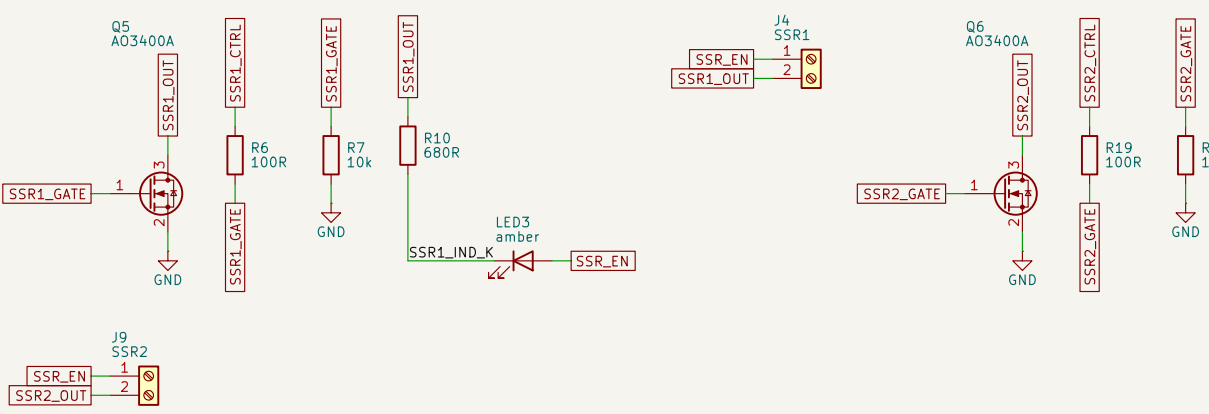
THERMOCOUPLE 1 (control) MAX31856  
T- floats to J3, biased via BIAS



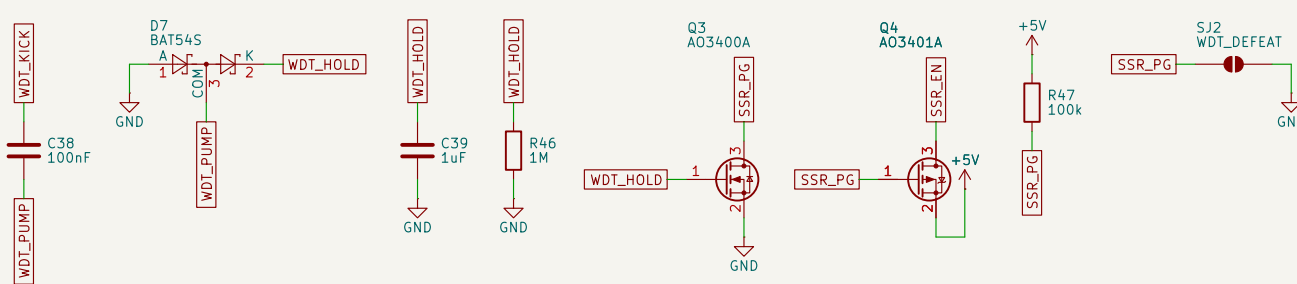
THERMOCOUPLE 2 (load) MAX31856  
T- floats to J8, biased via BIAS



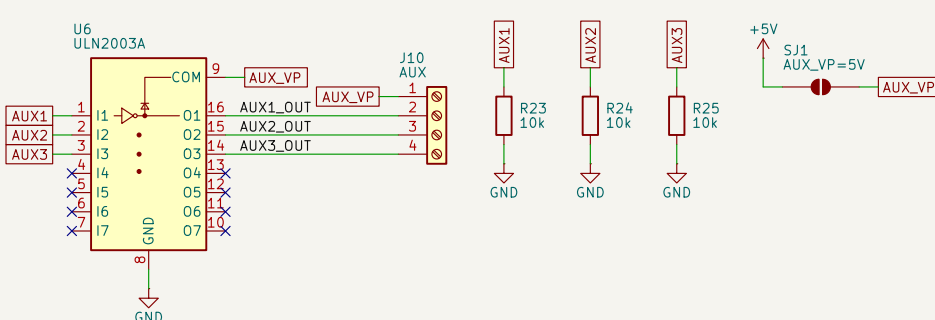
SSR DRIVE x2  
low-side AD3400A; indicator LED across each terminal



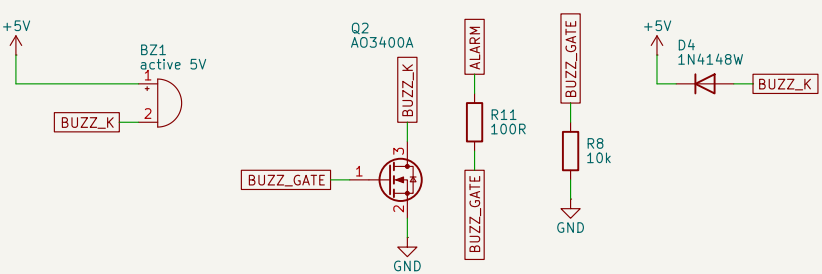
HARDWARE WATCHDOG  
C38/D7/C39/R46 diode charge pump on WDT\_KICK holds Q3 on; Q3 pulls SSR\_PG down, turning on high-side Q4, which supplies SSR\_EN - the +5V rail feeding BOTH SSR terminals and both indicators. R47 is the fail-safe pull-up. SJ2 = bring-up defeat, REMOVE for service.



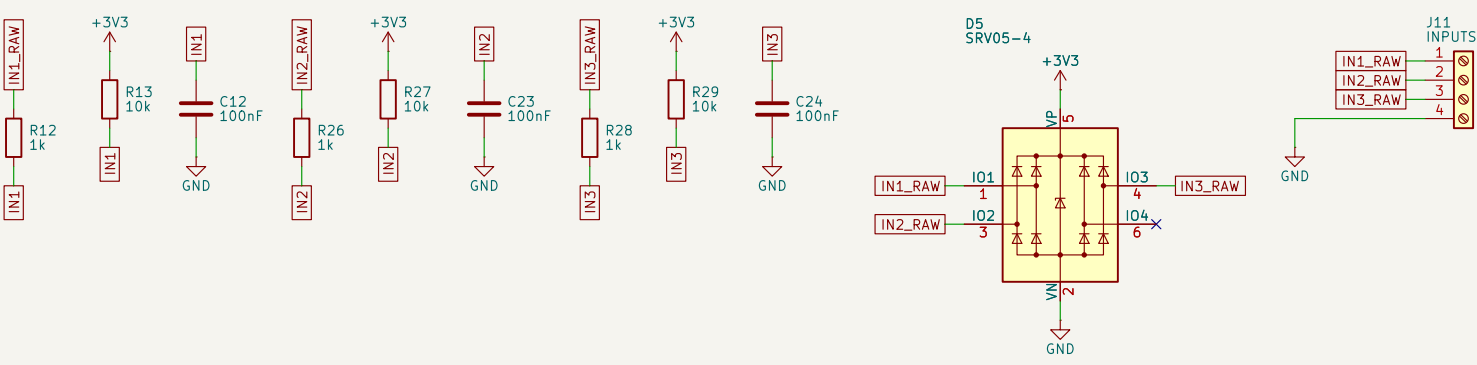
AUX OUTPUT BANK  
ULN2003 vent/purge/spare; COM->AUX\_VP, SJ1 links to +5V



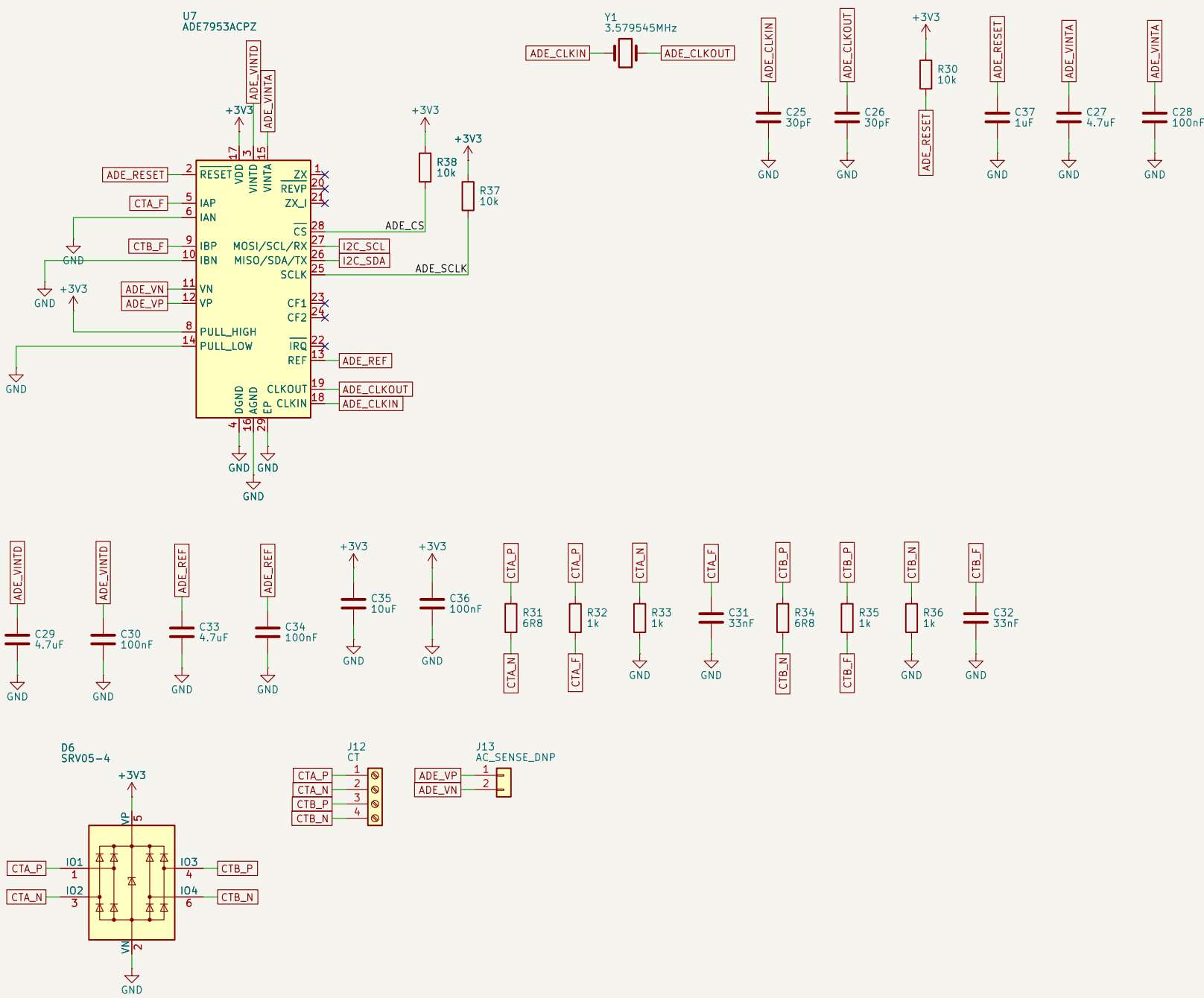
ALARM BUZZER



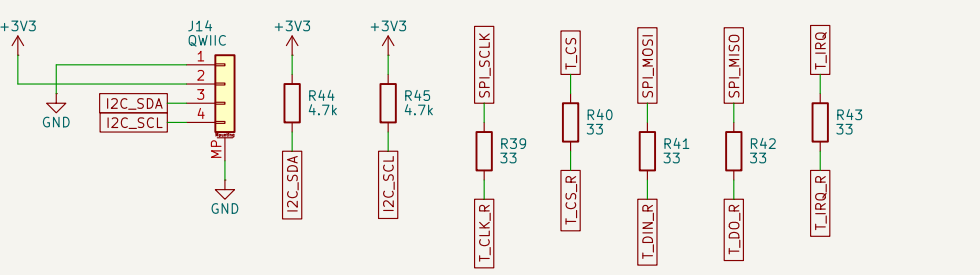
PROTECTED INPUTS x3 I<sub>lid</sub> / gas-flow / spare  
1k + 10k pull-up + 100nF each, SRV05-4 TVS



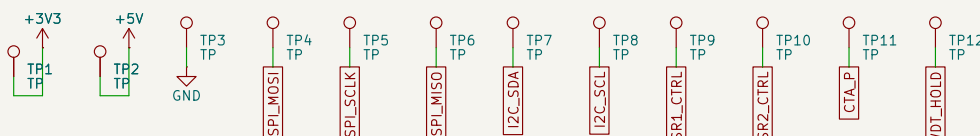
CT CURRENT SENSING ADE7953, I2C, current-only  
no mains - VP/VN to DNP J13



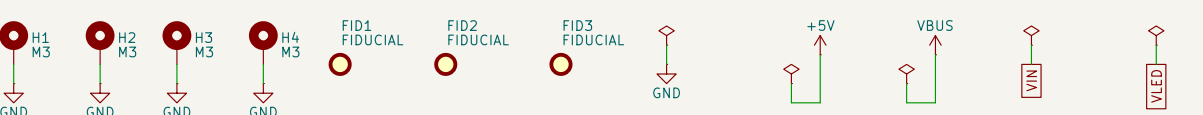
TOUCH DAMPING + I2C EXPANSION  
R39-R43 damp the shared SPI2 bus for the display module's XPT2046 (not on this board). J14 Qwlic and J7 5-8 (0.1 in) share the I2C bus, pulled up by R44/R45



TEST POINTS



MOUNTING / FIDUCIALS / POWER FLAGS  
FID1-3 are bare-copper pick-and-place targets, on no net



NOTES - Bisque kiln controller, rev B

SSR terminals J4/J9: pin1 = SSR\_EN (board +5V, watchdog-gated).  
pin2 = the switched low side, Boot-safe: R7/R20 hold each MOSFET gate down while the ESP32 pins are high-impedance.  
NOT opto-isolated - rev B tried that and reverted it: an opto only isolates if the SSR loop is powered off-board, and this board powers it.  
TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K- - both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required: two grounded-junction probes in one kiln would loop through it.

Nav switch J6 is panel-mounted: inputs use ESP32 pull-ups.

J11: IN1/IN2/IN3 (I<sub>lid</sub>/gas-flow/spare) + GND - dry contact each channel to GND.

Display J5 (14-pin, ST77965 + XPT2046 touch module, LCDWIKI MSP4021): 1=-5V 2=GND 3=CS 4=RS1 5=DC 6=SD0/MOSI 7=SC 8=BL 9=SD0/MISO 10=TD 11=TD 12=TD 13=TD 14=TD 15=TD 16=TD 17=TD 18=TD 19=TD 20=TD 21=TD 22=TD 23=TD 24=TD 25=TD 26=TD 27=TD 28=TD 29=TD 30=TD 31=TD 32=TD 33=TD 34=TD 35=TD 36=TD 37=TD 38=TD 39=TD 40=TD 41=TD 42=TD 43=TD 44=TD 45=TD 46=TD 47=TD 48=TD 49=TD 50=TD 51=TD 52=TD 53=TD 54=TD 55=TD 56=TD 57=TD 58=TD 59=TD 60=TD 61=TD 62=TD 63=TD 64=TD 65=TD 66=TD 67=TD 68=TD 69=TD 70=TD 71=TD 72=TD 73=TD 74=TD 75=TD 76=TD 77=TD 78=TD 79=TD 80=TD 81=TD 82=TD 83=TD 84=TD 85=TD 86=TD 87=TD 88=TD 89=TD 90=TD 91=TD 92=TD 93=TD 94=TD 95=TD 96=TD 97=TD 98=TD 99=TD 100=TD 101=TD 102=TD 103=TD 104=TD 105=TD 106=TD 107=TD 108=TD 109=TD 110=TD 111=TD 112=TD 113=TD 114=TD 115=TD 116=TD 117=TD 118=TD 119=TD 120=TD 121=TD 122=TD 123=TD 124=TD 125=TD 126=TD 127=TD 128=TD 129=TD 130=TD 131=TD 132=TD 133=TD 134=TD 135=TD 136=TD 137=TD 138=TD 139=TD 140=TD 141=TD 142=TD 143=TD 144=TD 145=TD 146=TD 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